How Much Fault Tolerance Is in a Fly Brain?

Fault-Injection Analysis of a Whole-Connectome Spiking Controller for Physical Intelligence

Shuchao Gao
Waseda University
Tokyo, Japan

***Abstract*—Whole-brain point-neuron models built from complete synaptic wiring diagrams have begun to serve as controllers in embodied settings; fault tolerance has been probed on subnetworks and via targeted silencings but not compared across hardware-motivated fault classes on a whole-brain controller. We take the released leaky integrate-and-fire (LIF) whole-brain model of *Drosophila melanogaster* (138,639 neurons, 15.1M signed connections, 54.5M synapse contacts) and inject five hardware-motivated fault classes—weight variability, connection loss, random-unit and hub output silencing, and sensory-channel deafferentation—scoring the population response of descending command neurons. Across 36 fault conditions (12 seeded trials each) under sustained escape-grade drive, the connectome absorbs 50% lognormal weight variability, retains 76% of command rate after 50% connection loss and under 10% random-unit output silencing, while silencing the top-1% hubs by total outgoing contact count produces a 2.5× global over-activity rather than silence. An edge-permuted null preserving weight and degree statistics yields a zero median command rate in every tested condition (no trial reaches the transmission threshold), and relative pathway decay is topology-independent: robustness is consistent with threshold amplification on convergence structure, not graph redundancy alone. We distill the results into five candidate design implications for fault-tolerant neuromorphic controllers.**

***Index Terms*—neuromorphic computing, connectome, fault tolerance, fault injection, Drosophila**

I. Introduction

PHYSICAL intelligence—agents whose perception, decision, and action loops close in the physical world under tight energy and latency budgets—demands controllers that remain robust on imperfect hardware [1], [2]. Neuromorphic systems promise the required efficiency [3], [4], and spiking implementations on Loihi 2 and FPGA substrates now cover vision and control workloads [5]–[7]. What remains poorly quantified is how *wiring* determines robustness: engineered spiking networks are designed task-by-task, so their fault tolerance must be re-measured per design [8], [9]. Complete signed wiring diagrams of whole brains are public, and point-neuron models built on them reproduce realistic dynamics without training [10], [11]; an evolved connectome is thus a field-tested controller and a reference for engineered fault tolerance.

The released leaky integrate-and-fire (LIF) whole-brain model of *Drosophila melanogaster* [10][1], which instantiates the complete FlyWire connectome [11], already used targeted computational silencings to identify circuit neurons [10], and network-dismantling work has deleted critical nodes from a derived sugar-sensing subnetwork [12]; we sweep five *hardware-motivated fault classes* on the fixed whole-brain model, scored jointly for transmission and activity-envelope compliance. We instrument the visuomotor escape pathway: looming-sensitive visual interneurons (LC4, LPLC2) converge onto descending command neurons (DNp01, DNa02, DNp04, DNb01) whose activation evokes take-off and escape steering [13]. Its failure modes map directly onto hardware fault classes: weight imprecision, broken interconnect, dead cores, dead sensors.

We inject five fault families (Section III-C), each over 12 seeded 600 ms trials, with two controls: an edge-permuted null preserving weights and degree statistics, and an analytic pathway-mass computation. Our contributions: (i) a fault-injection framework mapping hardware failure modes onto a connectome-constrained 138k-neuron LIF model; and (ii) tolerance curves of descending-neuron output in a complete connectome (50% weight noise absorbed, 50% connection loss retains 76%, 10% unit-output silencing leaves the median unchanged): top-1% hub silencing instead produces a $2.5\times$ *global over-activity* that passes transmission while violating the output envelope, and whole-channel deafferentation is the only catastrophic mode. We further (iii) show graph redundancy alone explains neither specificity nor graceful degradation (threshold amplification is a candidate account) and (iv) distill five candidate design implications.

II. Related Work

The release of complete wiring diagrams for the adult *Drosophila* brain [11] has shifted connectome research from static graph analysis to executable models: visual-system convolutions [14], neuron-level response prediction [15], learned tasks and whole-body locomotion on whole-CNS wiring [16]–[18], biofidelic processing units [19], and descending-neuron input sharing [20], [21]. Closest to our substrate, the released LIF whole-brain model (footnote 1) reproduces global activity and has been ported to Loihi 2 [5]. Dismantling studies have deleted critical nodes from connectome-derived circuits—including an SNN sugar-sensing subnetwork of the same released model, scored by a sensory firing-rate threshold [12]—but none sweeps hardware-motivated fault classes on a fixed whole-brain controller or separates output transmission from activity-envelope compliance; we do both.

[1]https://github.com/philshiu/Drosophila_brain_model

Graph-theoretic lesion studies score the structural attack tolerance of brain networks [22], not dynamics; spiking-network weight-noise robustness is mostly characterized on small trained architectures [8], [9], with adaptive mitigation now studied at the edge [23]; purpose-built fault-injection frameworks for trained SNNs—SpikeFI [24], SpikingJET [25]—automate this but score task accuracy on human-designed networks; follow-up work covers STDP-trained robustness and population-coded redundancy [26], [27]. Hardware platforms expose complementary constraints: on-chip variability co-design [6], FPGA precision–capacity trade-offs [7], permanent interconnect faults [28], edge energy–robustness envelopes [3], [29], and trained-CNN dependability [30]. Benchmarking efforts omit fault-injection axes [31], and formalizations of neuromorphic control do not yet treat degradation [32]. Insect-inspired collision-aversion circuits are robust by construction but small and hand-designed [33]. What a complete connectome uniquely enables is tolerance measurement at the scale of an entire evolved controller; the measurements suggest hypotheses for hardware validation.

III. Fault-Injection Framework

A. Whole-Brain Spiking Substrate

We adopt the released whole-brain LIF model of *Drosophila* [10] (Section I), implemented in Brian2 [34] and built on the FlyWire connectome [11]. Every neuron is a current-based LIF unit (threshold -45 mV, reset -52 mV, membrane time constant 20 ms, refractory 2.2 ms, axonal delay 1.8 ms). Each connection in the released table becomes a delta-conductance synapse whose jump is 0.275 mV per counted contact and decays with a 5 ms synaptic time constant, signed by the presynaptic cell's neurotransmitter; the table yields 15,091,983 connections between 138,639 neurons (54.5M contacts), 40% inhibitory. As in the released model, baseline firing is zero and the only external input is the stimulus group, so every recorded spike traces back to the stimulated cells. Our engine restores the full network state at the start of every trial, so all conditions share identical initial states, while the stimulus draws a fresh realization each trial: reported dispersion includes stimulus variability. One 600 ms trial runs in $\approx$3.5 s on one CPU core.

B. Task: Visuomotor Escape

The instrumented pathway (Fig. 1) is the fly's looming-triggered escape circuit: looming-sensitive visual projection neurons LC4 (104) and LPLC2 (210) converge on descending neurons [13], [33]. Because collision-sensing models treat looming as sustained high-rate drive, we drive the sensory cells (and, as a control, an equal-sized random set) with independent 150 Hz Poisson trains rather than modeling looming kinetics. The primary readout is the population rate of the command set {DNp01, DNa02, DNp04, DNb01} (8 neurons); all 1,299 descending neurons are a secondary readout. Calibration drove the command population to 94.6 Hz (interdecile 93.6–95.8 Hz, first-spike latency 4.4 ms; 10-trial set) under escape stimulation but only 8.2 Hz (6.4–9.2 Hz) under random stimulation; $\theta = 51.4$ Hz (the 94.6/8.2 Hz midpoint); every pass/fail classification below is unchanged for thresholds between the observed trial extremes (9.2–63.1 Hz); Section IV uses the 12 sweep trials.

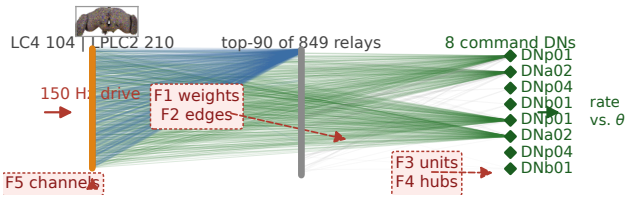


Fig. 1. The instrumented escape pathway, drawn from the connectome itself. Left: the 314 driven sensory cells (LC4, LPLC2); center: the 90 top-ranked of 849 two-hop relays; right: the 8 command DNs. Line width and opacity scale with synapse count (594 direct sensory→DN edges); dashed red marks the fault-injection points (F5 channels; F1/F2 synapses; F3/F4 units). Inset: reconstructed adult brain, adapted from Dorkenwald et al. [35] (CC BY 4.0).

C. Fault Taxonomy

Five fault families cover the hardware-relevant failure modes. F1 multiplies every synaptic weight by an i.i.d. mean-preserving lognormal factor ($\sigma \in \{0.05, 0.1, 0.2, 0.5\}$), modeling analog weight variability and bit-level imprecision. F2 deletes a random fraction $f \in \{0.05, 0.10, 0.25, 0.50\}$ of connections ($\approx f$ of contacts in expectation), modeling interconnect or memristor failures. F3 silences all outgoing connections of a random fraction $f \in \{0.01, 0.02, 0.05, 0.10\}$ of units, excluding input and readout cells (silenced cells still receive input and spike; only outputs are disconnected), modeling dead cores at the output boundary. F4 applies the same operation to the top 1%, 2%, or 5% of units ranked by total outgoing contact count (again excluding input and readout), modeling failure of high-fanout components; these cells carry 9–22% of all connections across the three doses, a targeted, correlated version of F3. F5 zeroes all outgoing connections of one (LC4 or LPLC2) or both sensory channels, modeling sensor or afferent-path failure. All faults are applied after state restoration; F4 and F5 act on fixed cell sets, while F1–F3 redraw their realization each trial from SHA-256 seeds over (phase, condition, trial)—12 independent draws per condition, exactly reproducible from the archived seeds (the stimulus stream is not re-seeded, so re-runs are statistically equivalent rather than bit-identical).[2]

D. Controls

Two controls isolate what the biological wiring contributes. The first is an edge-permuted null: every synapse keeps its presynaptic partner, weight, and sign, but its postsynaptic target is randomly reassigned. This preserves out-degrees, weights, and the global in-degree histogram while destroying which inputs converge onto each target—and with them all multi-hop structure; the identical sweep runs on it. The second is analytic: for both graphs we compute the *pathway mass* $m_k = \mathbf{1}_{\mathrm{DN}}^{\top} A^k \mathbf{1}_{\mathrm{in}}$, the synapse-weighted total of all k-hop

[2]Code, connectome-derived edge lists, and per-trial records: https://github.com/Scophield/drosophila-fault-injection.

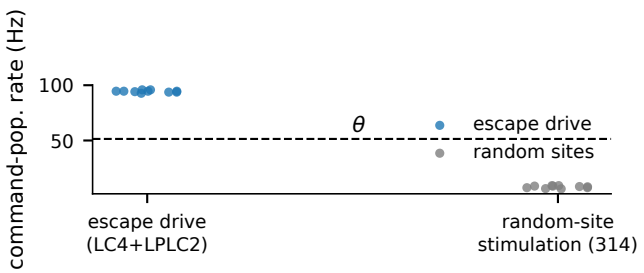


Fig. 2. Readout specificity. Command rate under escape stimulation (LC4+LPLC2) versus matched random-site stimulation (314 cells); dashed line: θ.

directed paths from LC4∪LPLC2 into the full descending layer (A holds contact counts, i.e., absolute signed strengths; m_k counts k-step walks of either sign—a wiring-density proxy, not signal flow), on the intact graph and after the F2 deletion levels; comparing its decay with the output decay separates topological redundancy from dynamical amplification.

IV. Results

A. Baseline: The Connectome Is a Specific, Low-Latency Controller

Driving LC4 and LPLC2 at 150 Hz produces a command rate of 94.7 Hz (median over 12 trials) with 4.05 ms first-spike latency, far above θ (Fig. 2). An equal-sized random set, driven identically, yields only 8.2 Hz on the same readout while producing *more* total activity (293k vs. 40k spikes): the readout is pathway-specific, not generic arousal. When both sensory channels are silenced ($F5_{\text{both}}$), the command rate collapses to zero in every trial; the residual 21.2k spikes are, by construction, the stimulated sensory cells themselves (the rest of the brain starts at zero baseline and receives no input), so the score measures transmission through the instrumented pathway.

B. Tolerance Curves: Noise Is Absorbed, Disconnection Degrades Gracefully

Figure 3 shows dose–response curves for the two most consequential families; F1 and F3 responses are flat in median (Table I). Weight variability (F1) leaves the median unchanged: at σ=0.5 the rate is 94.2 Hz (healthy: 94.7 Hz), the trial dispersion widens (interdecile 90.4–100.8 vs. 94.0–95.6 Hz), and total activity rises only from 39.7k to 41.6k spikes. Random connection deletion (F2) degrades gracefully: 5% and 10% loss leave the rate at 91.9 and 93.1 Hz, 25% loss retains 84.5 Hz (89.2% of baseline), and even 50% loss retains 72.4 Hz (76.5%), with all 12 trials passing transmission in every condition (weakest 63.1 and 68.1 Hz, above θ by ≥22%). Disconnection's cost appears predominantly in latency: 4.05 to 5.4 ms at 50% loss (+33%), consistent with weakened aggregate drive rather than longer residual routes alone; which dominates a task-level error budget is workload-dependent. Random unit silencing (F3) up to 10%—nearly 14,000 neurons cut off from their targets—leaves the median unchanged (95.1 vs. 94.7 Hz): command drive integrates over hundreds of converging populations, and no tested random slice is load-bearing. The wide 1,299-neuron descending layer tracks the same pattern (2,941 spikes healthy vs. 1,918 at 50% loss), so the graded response is a property of command encoding, not the eight-neuron readout. Aggregation masks composition: four of the eight command cells carry 96.6% of healthy readout spikes, and at 50% connection loss the smallest contributors collapse (158→10, 26→0) while silent cells are recruited (1→58, 0→21); the curves are a descending-population response, not behavior.

C. Hub Silencing Produces Over-Activity, Not Silence

The dose–response of hub silencing (F4) is non-monotonic and qualitatively different from every distributed fault (Figs. 3b, 4b). Silencing the top 1% of cells by outgoing contact count (1,386 neurons, 9.1% of connections) *increases* the command rate to 122.3 Hz (+29%, interdecile 121.1–123.9 Hz across trials) and more than doubles global activity (99.9k spikes, 2.52× healthy; Mann–Whitney vs. healthy $p < 10^{-4}$, as for $F2_{50}$ and $F5_{\text{both}}$); at 2% the rate stays elevated (117.1 Hz, activity 42.4k); at 5% (22.4% of connections) both metrics return to near-baseline (94.2 Hz, 36.5k). A matched-disconnection contrast suggests the effect is not disconnection alone: random 10% unit silencing disconnects 10.0% of connections (more than the hub set's 9.1%) yet leaves the readout unchanged (95.1 Hz); a control matching cell count and excitation/inhibition composition is still needed. The removed hub output is inhibitory-rich (62% vs. 40% network-wide), making de-inhibition the leading hypothesis for the over-drive. We therefore score *transmission* (rate ≥ θ) and an *activity envelope* (rate and total activity within 2× of healthy) separately: hub silencing at 1% passes transmission while violating the envelope in all 12 trials (persists for multipliers 1.5×–2×); every distributed fault passes both; $F5_{\text{both}}$ fails both. The envelope violation is the dangerous mode: output exceeds its envelope while every per-unit monitor reads "active." The same silencing in the permuted null produces no such signature, tying the over-activity to biological hub structure.

D. What the Wiring Contributes: Null Network and Pathway Analysis

The permuted null dissociates specificity from redundancy: out-degrees, weights, and the in-degree histogram survive, convergence is reassigned, and its median command rate is 0.0 Hz in every tested condition, with no trial reaching θ; its broad 1,299-neuron readout falls to background (45.5 spikes per trial vs. 2,941 healthy) and global activity to about 60% of the biological level (23.8k vs. 39.7k)—little more than the 21.2k the driven cells produce with all outputs severed ($F5_{\text{both}}$): the null receives the same drive, but only ∼2.6k spikes propagate, and the descending readout stays at background. Pathway mass quantifies the convergence advantage: 5.8× (1-hop) and 17.7× (4-hop) more route mass into the descending layer. Yet *relative* pathway-mass decay is nearly

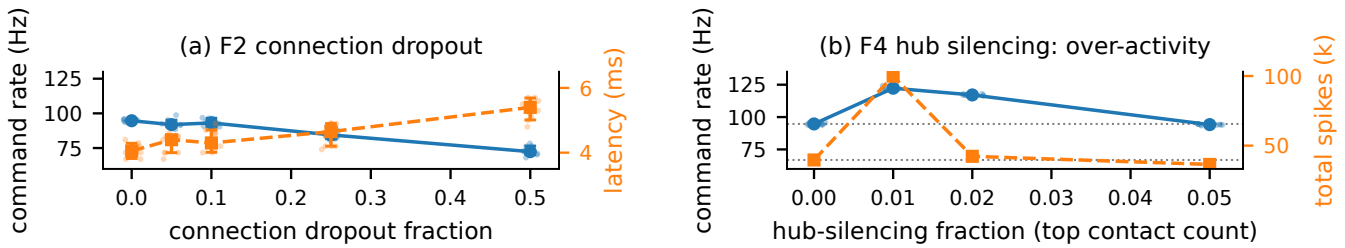


Fig. 3. Tolerance curves of descending-neuron output (medians over 12 trials; blue: rate; orange: latency (a) / total spikes (b); dotted grey: healthy medians). (a) Connection dropout: 76% of command rate retained at 50% loss, latency +33%. (b) Hub silencing: over-activity.

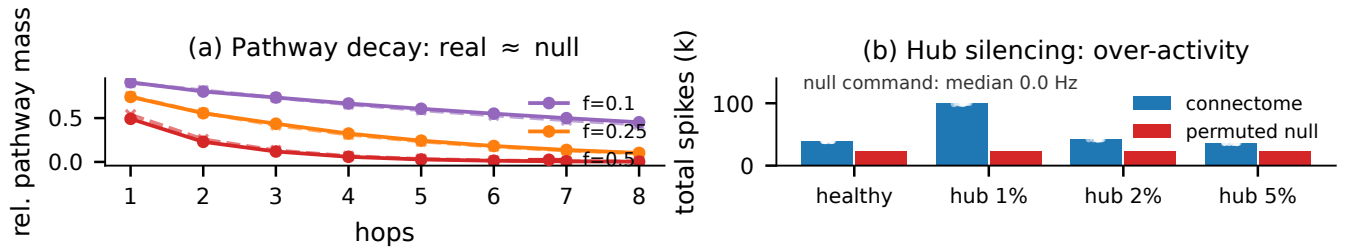


Fig. 4. Topology contrast. (a) Route mass under edge loss, normalized per graph (connectome solid, null dashed): relative decay nearly identical, absolute mass 5.8–17.7× higher. (b) Activity under hub silencing: over-activity only in biological wiring; null readout 0.0 Hz (median).

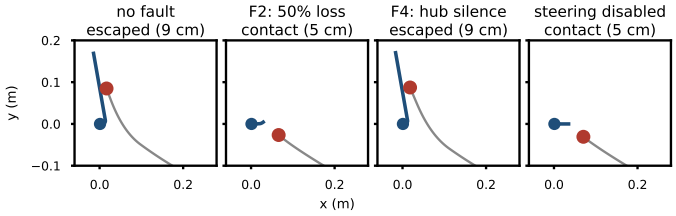


Fig. 5. Synthetic closed-loop escape at −30° (seed 0). Blue: agent; grey: threat; red: endpoint. Panels: no fault, $F2_{50}$, $F4_{1\%}$, steering disabled. Success: minimum separation > 0.05 m.

identical (at f=0.5, k=8: 0.3% vs. 0.4% of baseline; Fig. 4a), while output degrades far more slowly (76% at f=0.5 against 5.8% four-hop route mass)—ruling out graph redundancy alone as the mechanism. A candidate account is threshold amplification: with N converging afferents and per-synapse survival p, aggregate drive scales as Np while k-hop route mass decays as p^k in expectation (6.25% at k=4, p=0.5), and the saturating threshold compresses drive changes into small rate changes. Retention above proportional scaling (76.5% at p=0.5, 89.2% at p=0.75) is consistent with this account but does not establish it: in an excitatory–inhibitory network even linear signal flow need not scale with p, since deleting inhibitory connections offsets excitatory loss. Disentangling these requires E/I-balanced deletion and drive-strength sweeps.

E. From Transmission to Behavior: A Closed-Loop Escape Demo

Transmission retention need not imply task capability, so we close the loop synthetically: per-cell looming rates over the 314 instrumented LC4/LPLC2 cells (assumed receptive fields), the full network in the loop, and a calibrated left–right difference across the eight command cells—trained on two azimuths, then frozen—steering a planar agent away from a pursuing threat, with a ballistic takeoff triggered by the 8-cell *average* rate against a scene-calibrated threshold (θ_{demo}=26.3 Hz, the training trajectories' median early-response level; the open-loop θ=51.4 Hz was calibrated for full-pool drive and does not transfer). It escapes 24/24 at four azimuths (six seeds; median miss 0.09–0.11 m vs. 0.05 m contact), 12/12 at held-out azimuths, 0/3 frontal (the left–right blind spot; Fig. 5). Every control collapses (0/12: steering, both branches, reversed sides); a scene-reading hand rule is an upper bound (12/12). Faults transfer with a threshold cliff: 25% loss gives 10/12, 50% gives 0/12 (readout rate falls below θ_{demo}), and hub silencing preserves escape (12/12) while raising activity to 1.8×—the ordering seen in open loop. This is a preliminary synthetic demonstration, not a behavioral validation (assumed encoding, unmodeled gaze, scripted loom envelope, kinematic agent).

V. Candidate Design Implications for Embodied Neuromorphic Controllers

P1 and P2 follow from the flat F1 and F3 medians: computation lives in distributed convergence, consistent with hardware observations [6], [8]. P3: disconnection costs are latency; real-time loops should budget Fig. 3a. P4: hub silencing is the only *amplifying* mode; monitor envelope violation, not liveness. P5: each sensory channel alone suffices (68–79% median), marking the sensor interface as the first redundancy candidate.

TABLE I
CANDIDATE DESIGN IMPLICATIONS FROM CONNECTOME FAULT INJECTION, WITH QUANTITATIVE ANCHORS.

#	Implication	Anchor (this study)
P1	Redundancy of wiring, not precision of weights	σ=0.5 lognormal noise: 99.4% retention
P2	Distribute command integration	10% unit-output silencing: 100.4% retention
P3	Budget latency for disconnection faults	50% connection loss: 76% rate, +33% latency
P4	Regulate high-fanout units; monitor envelope	top-1% hub silencing: 2.5× over-activity
P5	Harden the sensory interface first	single-channel silencing retains 68–79% median; both 0%

Identical relative pathway decay yet only the biological graph transmitting a decodable command, argues against certifying embodied controllers by topology alone; fault-injection campaigns belong in pre-deployment stages [31], [32].

VI. LIMITATIONS AND CONCLUSION

Our LIF abstraction (single compartments, no plasticity, Poisson drive) fixes one drive rate; doses kept the background regime intact, the hub analysis is descriptive, and correlated damage and recovery are excluded. Under the tested drive, the median command rate is preserved under weight noise and unit silencing and retains 76.5% at 50% loss; the dangerous failure is hub-silencing over-activity, an ordering a preliminary synthetic loop reproduces.

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